

Math Camp 2026 — Day 2 Exercises

Calculus · Linear algebra · Data wrangling

Calculus

1. Differentiate $f(x) = 4x^3 - 2x + 7$.
2. Differentiate, using the chain rule:
 - (a) $g(x) = e^{3x}$
 - (b) $h(x) = \ln(x^2 + 1)$
3. Let $f(x) = -2x^2 + 8x - 3$.
 - (a) Find the critical point with the first-order condition.
 - (b) Use the second-order condition to say whether it is a maximum or a minimum.
4. Let $f(x, y) = 3x^2y + y^3 - 5x$.
 - (a) Find $\partial f / \partial x$ and $\partial f / \partial y$.
 - (b) In one sentence: what is being held constant in part (a)?
5. A random variable X has a flat density on the interval $[0, 4]$ — that is, $f(x) = c$ for $0 \leq x \leq 4$ and 0 elsewhere.
 - (a) What must c be?
 - (b) What is $P(1 \leq X \leq 2)$?

Linear algebra

6. A is 3×4 and B is 4×2 .
 - (a) Is AB defined? If so, what are its dimensions?
 - (b) Is BA defined?
7. You run a regression on 150 countries with an intercept and 4 covariates.
 - (a) Give the dimensions of y , X , and β in $y = X\beta + \varepsilon$.
 - (b) Name one thing you could do to X that would make $(X'X)^{-1}$ fail to exist.

Data wrangling

8. Rewrite this using the pipe:

```
summarize(group_by(filter(df, pop > 1e6), region), mean_gdp = mean(gdp, na.rm = TRUE))
```

9. x has 120 rows. After `left_join(x, y, by = "iso3")` you have 137 rows.
 - (a) What went wrong?
 - (b) Which join would help you find rows of x that matched nothing in y ?
10. You have this table:

country	trade_2019	trade_2020
USA	4.1	3.8
JPN	1.5	1.4

- (a) Which tidy-data rule does it break, and why?

(b) Write the `pivot_longer()` call that fixes it.